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PALAEOMAGNETISM OF LATE PRECAMBRIAN DOLERITE DYKES FROM VARANGER PENINSULA, NORTH NORWAY

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Palaeomagnetic investigations have been carried out on 12 dykes of Late Precambrian age from the Varanger peninsula, north Norway. The dykes are separated into two groups, the Kongsfjord dykes and the Båtsfjord dykes. In the Kongsfjord dykes, titanomagnetite is almost entirely erased, as a result of an extreme degree of alteration. Pyrrhotite is the dominating magnetic mineral, and only three stable specimen directions can be defined. In the Båtsfjord dykes, however, the most important magnetic constituent is nearly pure magnetite, and a two-axis magnetization structure is revealed. The directions of the major component conform to a Fisherian distribution, and are assumed to represent the relative Late Precambrian field. Superimposed on this magnetization is a minor component which is assumed to be of Caledonian origin, probably Ordovician. This latter remanence is in accordance with other Middle Palaeozoic results obtained in Western Europe. The upper age limit of the Late Precambrian field is discussed, and it is proposed that the polar shift from the Late Precambrian position to the main Palaeozoic group may have occurred as late as Middle Ordovician.

1. Introduction

The current discussion among palaeomagnetists which has extended over the last decade on the interpretation of Palaeozoic and Precambrian palaeomagnetism, has shown that proper understanding of such data may be exceedingly problematic. Firstly, age determinations of such old rocks are often uncertain; fossils are rare, and radiometric dating methods are sensitive to alteration processes that have often taken place during the history of the rocks. Secondly, the magnetization is generally multi-component, and therefore there are problems with regard to isolating the various sub-components of magnetization and in determining their chronological order. However, it has been repeatedly pointed out by Storetvedt (1973, 1978), Storetvedt and Markhus (1978) and others that careful examinations of the directional trends may in many cases provide valuable information about the magnetization history. Also, the vector-subtraction method as used by Roy and Park (1974) has been shown to be a useful tool for dealing with multivectorial magnetization.

The present paper reports on Late Precambrian and Early Palaeozoic palaeomagnetic properties investigated in dolerite dykes from the Varanger peninsula, north Norway. Through a detailed directional trend analysis, combining vector subtraction and rock-magnetic studies, determination of the various magnetization components and their origins has been attempted. Megatectonic implications of the results have been dealt with elsewhere (Kjøde et al., 1978).

2. Geology and sampling

The Varanger peninsula of Finnmark, north Norway (Fig. 1), is divided into two separate geologi-

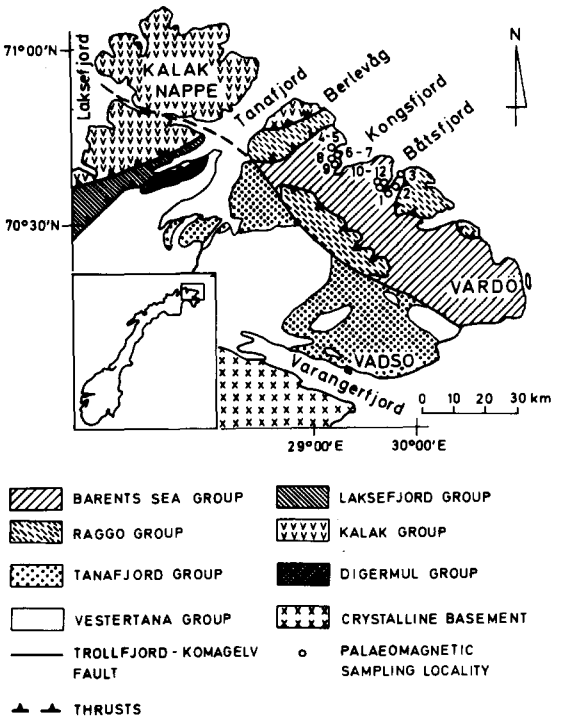


Fig. 1. Geological sketch map of the Varanger peninsula and adjacent areas showing site locations.

cal regions by a major tectonic dislocation, the NW–SE trending Trollfjord–Komagelv (T–K) fault zone (Siedlecka and Siedlecki, 1967). Southwest of the fault, an autochthonous sedimentary succession, ranging from Upper Riphean to Tremadocian, rests unconformably on a high-grade crystalline basement. Northeast of the fault zone, there is an allochthonous sedimentary sequence of Riphean age, including the 9000-m thick Barents Sea Group (cf. Fig. 1; Siedlecka and Siedlecki, 1967, 1971, 1972; Banks et al., 1971; Siedlecka, 1975). This group is divided into Kongsfjord (oldest), Båsnæring, Båtsfjord and Tyvjofjell formations (Siedlecka and Siedlecki, 1971, 1972), and is dissected by numerous nearly vertical NE-trending dolerite dykes: these intrusions may locally attain swarm proportions (Holtedahl, 1918). Tectonic analysis (Roberts, 1972) leads to the conclusion that northeast of the T–K Fault there has been one main fold-producing deformation episode (D_1). Most of the dykes seem to be coeval with this event, being parallel (or nearly so) to the fold axial surface, and carrying the same axial planar cleavage which pervades the host sediment piles. Roberts (1975) describes these dykes as metadolerites, to distinguish them from younger, uncleaved, and post-folding dolerites. The metadolerites occur exclusively northeast of the T–K dislocation and most frequently in the Kongsfjord and Båsnæring formations; while the far less common uncleaved dykes are found on both sides of the fault, although largely restricted to the south-eastern district of the peninsula.

According to K/Ar age determinations, Beckinsale et al. (1976) conclude that there are three generations of basic dykes on the Varanger peninsula. The uncleaved dykes suggest an average age of about 355 My B.P., whereas data from the metadolerites show inconclusive and unreliable ages, ranging from 935 to 1946 My B.P.; this uncertainty is probably due to their low K content, which may result in excessive age errors. On the other hand, a series of uncleaved or weakly-cleaved dykes, cutting through the Båtsfjord formation, give well-defined ages of around 640 My B.P. The geochemical and petrogenetic analysis of Roberts (1975) suggests that the latter dykes (here named Båtsfjord dykes) are very similar to the metadolerites (hereafter referred to as Kongsfjord dykes), and that most likely they all belong to the same 640 My B.P. intrusive phase (during the D_1 deformation).

A total of 80 hand samples were collected from 12 dykes: 6 dykes from each of the Kongsfjord and Båtsfjord suites (cf. Fig. 1). In the field, alterations in most of the dykes were easily observable; especially in the Kongsfjord region where veinlets of quartz, chlorite and calcite, along with fracturing and cleavage development, were frequently seen. The dykes were of a green and brownish appearance, although the Båtsfjord dykes are generally rather fresh-looking compared to those from Kongsfjord. Blocks were taken from all parts across the dykes, and both magnetic- and sun-compasses were used for orientation. The thickness of the dykes varied between 3 and 5 m. All the dykes showed clear, baked contact zones with the sediments. An attempt was made to apply the baked contact test to samples collected from the country rock: however, the magnetization was too weak to be measured on the available instruments.

3. Magnetomineralogy

3.1. Microscopy

Polished sections of 15 specimens from the Båtsfjord dykes, and 13 from the Kongsfjord dykes, have been examined under reflected light at magnifications of up to 2000 X. The magnetic colloid technique has been used for identification purposes, and to detect magnetic structures within the magnetic grains.

In the Båtsfjord specimens, more than 75% of the opaques are titanomagnetite and its alteration products. The other minerals are free ilmenite and sulphide (pyrite). There is more than 5% by vol. of opaque minerals in these rocks. The titanomagnetite grain sizes range up to 1000 μm, the average size being ca. 100 μm. Broad ilmenite exsolution lamellae cut through some of these grains, and the further alteration of ilmenite into pseudobrookite suggests that at least some of the grains can be placed in the high-temperature oxidation classes IV and V (Ade-Hall et al., 1968). However, the most predominant feature observed is strong low-temperature oxidation (maghemitization) and granulation. Many grains are more strongly magnetic along the edges than in the interior (as revealed by the magnetic colloid technique). In addition, it appears generally that the grains have been exposed to an extreme degree of fine-scale distintegration. However, in some of the most disintegrated grains the ilmenite lamellae still remain unaltered. This indicates that the temperature during the low-temperature processes was in general below 250°C (Ade-Hall et al., 1971).

In the Kongsfjord dykes, the titanomagnetites are almost completely obliterated. The presence of titanomagnetite grains is, in contrast to the above, largely restricted to relics of titanomagnetite, although lamellae of altered ilmenite are still found. The predominating opaque minerals are sulphides, and the main magnetic constituent appears to be pyrrhotite. Grain sizes range from less than 5 μm to greater than 1500 μm. In these dykes the opaque minerals constitute less than 2% by vol., which is a considerably smaller proportion than in the Båtsfjord dykes.

3.2. Thermomagnetic analysis

Strong-field (3.3–6.5 kG) thermomagnetic curves have been determined on 39 specimens from the Båtsfjord dykes, and 25 specimens from Kongsfjord dykes, using a horizontal-beam Curie balance. Heating (to max. 700°C) and subsequent cooling have been carried out in air, at a rate of 20–35°C min^{-1}. The grain sizes of chip specimens were usually in the range 0.5–3.0 mm, while powdered specimens with grain sizes < 50 μm have been used to study oxidation effects. Typical J_s/T curves for the Båtsfjord and Kongsfjord dykes are shown in Fig. 2a and b respectively.

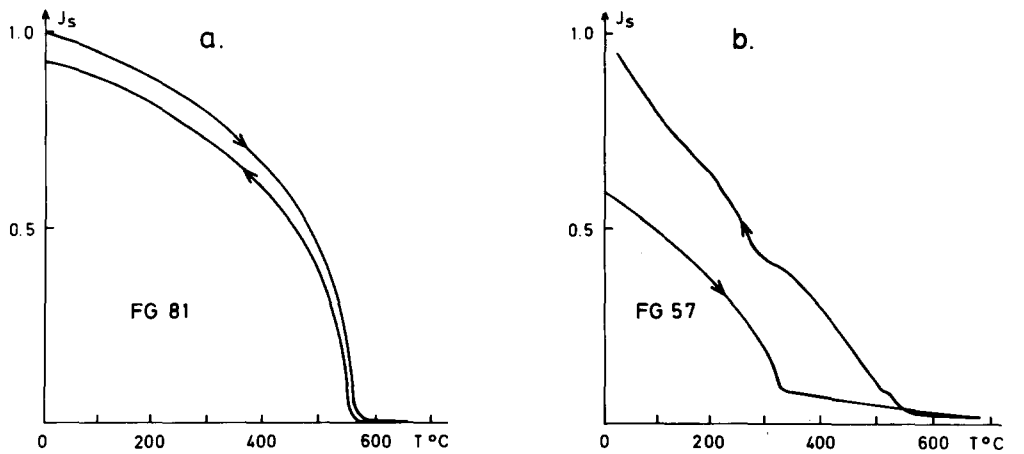


Fig. 2. Representative examples of thermomagnetic curves from (a) the Båtsfjord dykes, and (b) the Kongsfjord dykes.

On heating, the Båtsfjord rocks show a single, distinct T_C between 570 and 590°C. In the majority of these specimens, the J_s/T curves are fairly reversible, although after heating there is generally a minor decrease in J_s at room-temperature. This decrease can be explained either by the oxidation of magnetite, or by the inversion of titanomaghemite to haematite. However, experiments with powdered specimens suggest that oxidation is the dominant process. The Curie temperature of about 580°C indicates that the most important magnetic constituent of the Båtsfjord dykes is nearly-pure magnetite.

The Kongsfjord specimens produce a distinct Curie phase between 325 and 400°C. This is a typical T_C for pyrrhotite (Haraldsen, 1937) which, according to microscopic evidence, is the principal magnetic constituent of these dykes. Some heating curves also show an additional inflexion temperature around 550°C. According to Bhimasankaram and Lewis (1966), pyrrhotite has characteristic Curie temperatures corresponding to these two intervals. However, Cornwell (1967) found that the higher inflexion temperature was due to magnetite produced from pyrrhotite during heating. The portions of the thermomagnetic curves for cooling are characterized by a strong increase of the 550°C component, with a corresponding decrease (almost leading to disappearance) of the lower Curie phase. Preheating polished sections of a specimen to 700°C shows that the 550°C component is likely to be due to magnetite; this conclusion is also consistent with the increase in J_s exhibited after cooling.

In conclusion, there is good correspondence between the thermomagnetic and microscopy studies. It appears likely that the difference in magnetic mineralogy between the Båtsfjord and Kongsfjord dykes reflects a difference in the extent of secondary alteration.

4. Remanence characteristics

From each sample, $\frac{3}{4}$-inch cylindrical specimens were cut, and in general the remanence parameters were measured on a Foster spinner-magnetometer (Foster, 1966). However, for the most weakly magnetized rocks, a long-periodic astatic magnetometer was used (giving reliable measurements down to intensities of ca. 3×10^{-7} e.m.u. cm^{-3}). Standard procedures of thermal and alternating-field (AF) demagnetization have been used. The vector-subtraction technique (Roy and Park, 1974) has been applied to specimens following great-circle paths on progressive demagnetization.

4.1. The Båtsfjord dykes

The NRM intensities of the Båtsfjord specimens are of the order of 10^{-3}–10^{-4} e.m.u. cm^{-3}.

AF demagnetization was performed on 82 specimens, from 40 samples. The majority of the specimens are dominated by low coercivity forces (Fig. 3). Magnetization resistant to field strengths above 1000 Oe is found in only 6 specimens, but in these cases also the low coercive forces contribute a greater part of the NRM moment (cf. FG 19-C3 in Fig. 3.).

Representative results of the thermal treatment of 92 specimens are shown in Figs. 4 and 5. On the whole, the most marked blocking-temperature range is between 450 and 550°C (73 specimens). Above this temperature region, the intensity decay curves either flatten out, perhaps with a minor increase in the intensity, or show erratic behaviour. In 18 specimens, more than 80% of the NRM moment remains before the final intensity decline; but in most cases, a linear intensity decrease to less than 50% appears below the higher blocking temperature. Some specimens have a linear decay pattern throughout the range of demagnetization.

A main direction of magnetization ($D_m = 86°$, $I_m = +60°$) is defined by 74 specimens after partial thermal- or AF-demagnetization. This direction component is randomized below 300 Oe in most specimens by AF cleaning. However, in the few cases where the magnetization can withstand field strengths above 1000 Oe, the direction is also connected with the high-stability coercive forces (Fig. 3). The main direction of magnetization is associated, by thermal demagnetization, with blocking temperatures ranging upwards to 520°C (Figs. 4 and 5), and may contribute more than 80% of the NRM intensity.

In addition to the major component, three minor remanence vectors have been detected. Two of these components are the antiparallel magnetizations of a practically horizontal NNE–SSW-striking palaeomagnetic axis. Although the latter axis is not well defined

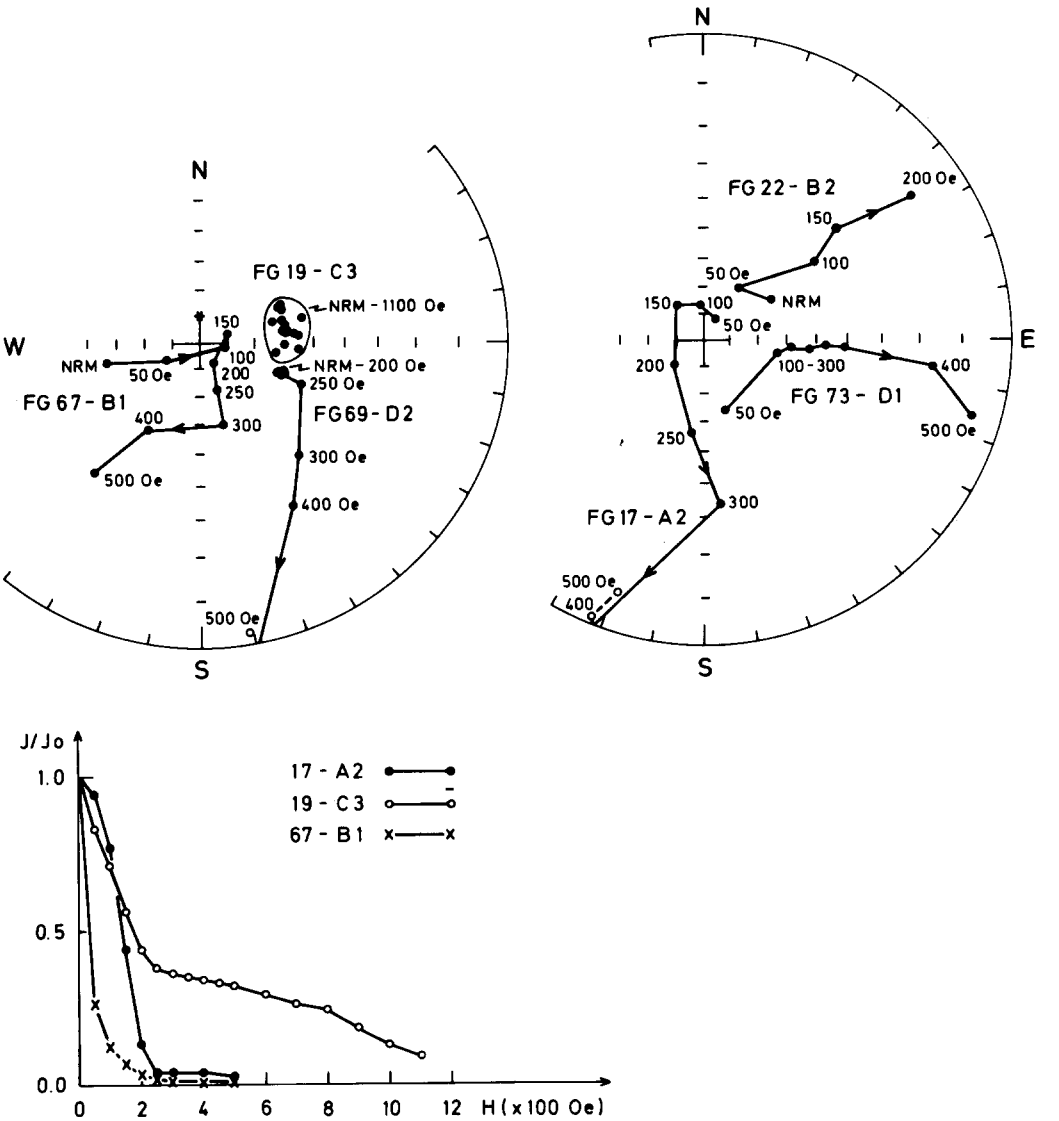


Fig. 3. AF-demagnetization results of specimens from the Båtsfjord dykes. Closed symbols, positive; open symbols, negative inclination in stereographic projection. Normalized intensity curves. Star: present axial dipole field direction.

by many stable specimen directions, there is sufficient experimental evidence to support its existence. The orientation of the axis is tentatively defined by the stable remanence vectors for specimens FG 24-H and FG 1-B1 (Fig. 4), and by the high-temperature "terminal" directions of specimens FG 2-B2 and FG 12-D2 (Fig. 5). This shallow magnetization axis is compatible with the large number of directional swings (at an advanced stage of demagnetization) towards NNE or SSW. Most of these trends occur after removal of more than 80% of the NRM moment on both thermal and AF demagnetization. Unfortunately, the erratic stage of demagnetization (not shown in the diagrams) occurs before these high-stability directions are cleanly separated. However, as these components may constitute 15–20% of the NRM intensity, they may well influence the direction of the bulk component (at least in some specimens).

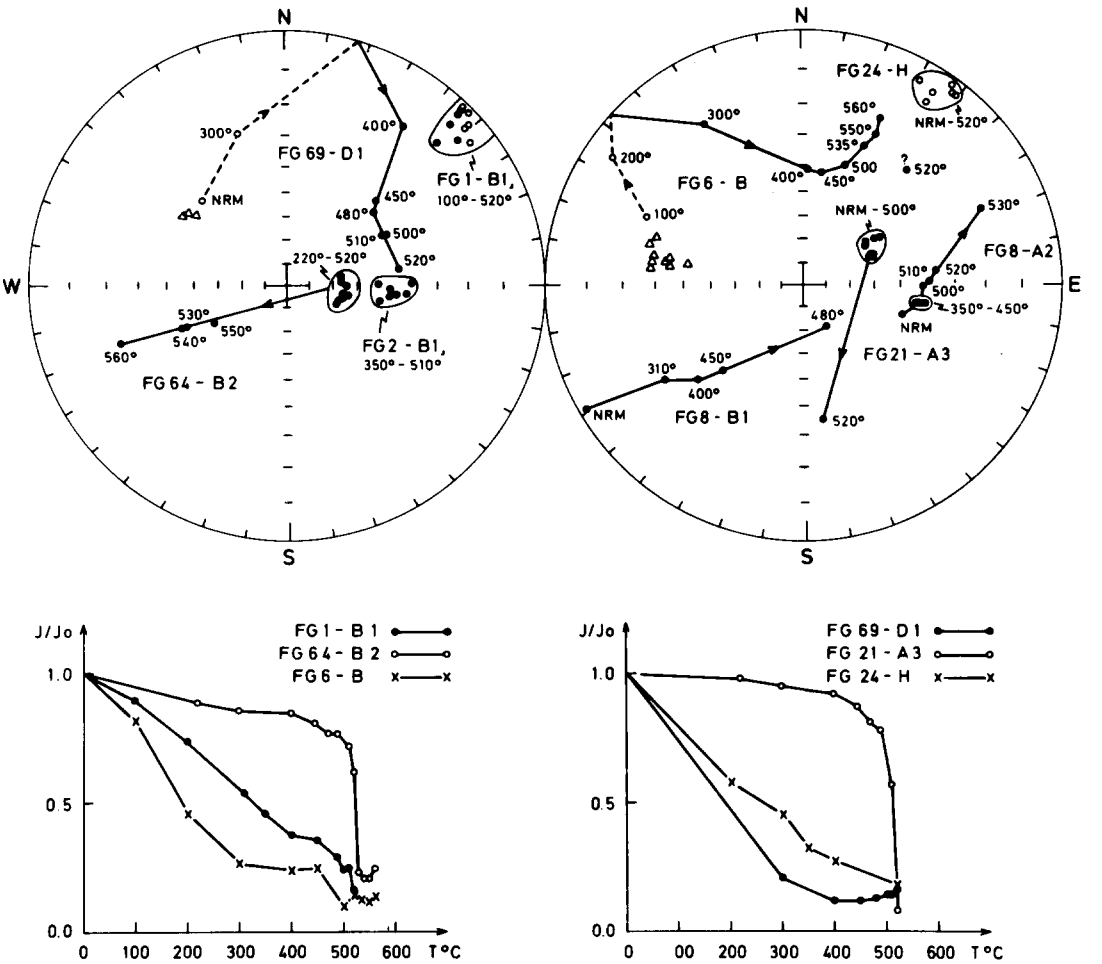


Fig. 4. Thermal-demagnetization results for the Båtsfjord dykes. Triangles represent directions after vector subtraction. For further explanation see the text.

To investigate the significance of this effect, vector analysis has been performed on a couple of specimens in which these components are assumed to give a maximum deflection of the estimated bulk direction. The maximum northeastward shift of the bulk direction is estimated, since the corresponding southwestward directional trends appear to be twice as frequent (35 specimens) as the northeasterly ones (16 specimens). Therefore, if any correction to the overall bulk direction should be necessary, a northeastward shift is expected.

The effect of vector subtraction is shown in Fig. 5. For FG 12-D2, it is suggested that the vector $\mathbf{R}_{535}$ is an end-point vector, and so this vector is subtracted from the two-component vectors in the range 100–520°C. Equivalently, the $\mathbf{R}_{560}$ vector of FG 2-B2 has been subtracted from the corresponding vectors in the range NRM–450°C. The vector-subtracted directions show a 20–25° northeastward shift relative to the bulk directions. Taking into account that (a) about half of this northeastward deflection will be compensated by specimens with opposite direction trends, and (b) that only a few specimens are likely to provide a deflection as large as 20–25°C, we can probably narrow down the directional correction for the main component to less than 10°. Since this is a minor adjustment without any direct significance for the later discussion it will not be considered further.

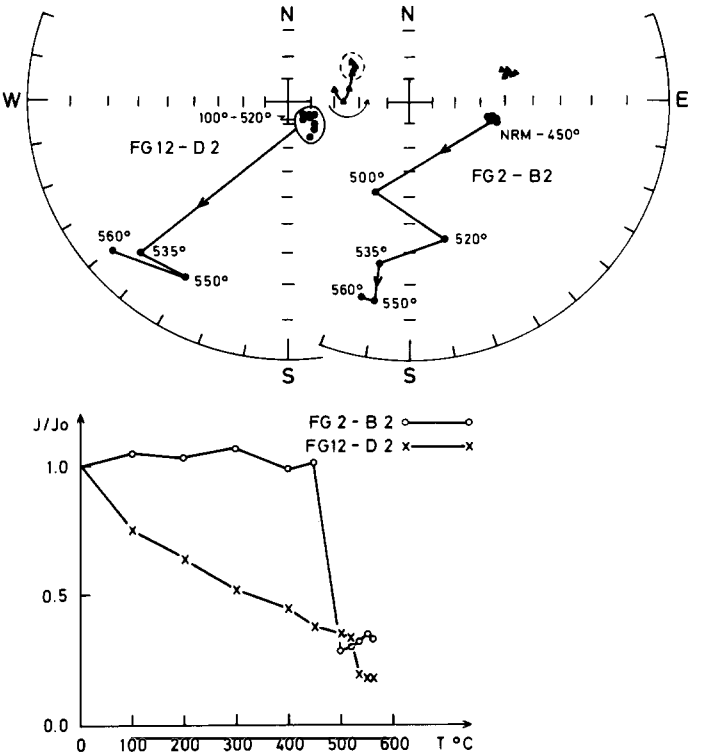


Fig. 5. Thermal-demagnetization results for the Båtsfjord dykes. Triangles represent directions after vector subtraction. For further explanation see the text.

Two of the thermally demagnetized specimens, FG 69-D1 and FG 6-B (Fig. 4), bear indications of containing the two antiparallel vectors of the main palaeomagnetic (E–W) axis. Both specimens show a systematic change in remanence direction (away from a westerly, moderately steep, upward-pointing position), following an approximate great-circle path. The most rapid variations in intensity and direction occur below approximately 400°C. Vector-subtraction analysis has been performed in an attempt to uncover the magnetization direction of the least stable component. For both specimens, the various difference-vector estimates (given by triangles in Fig. 5) for temperatures $< 400°C$ (for example, $\mathbf{R}_{NRM} - \mathbf{R}_{300}$, $\mathbf{R}_{NRM} - \mathbf{R}_{400}$, $\mathbf{R}_{300} - \mathbf{R}_{400}$, for specimen FG 69-D1) agree very well and define approximate opposite directions to the bulk magnetization. It is suggested therefore that the vector-subtracted directions represent true palaeomagnetic field markers.

It can thus be concluded that the Båtsfjord dykes tend to have as many as four different sub-components of magnetization, aligned along two entirely different palaeomagnetic axes.

Statistical analysis has been performed on the distribution of the major magnetization component of these dykes. Computer graphics of the distribution are shown in Fig. 6. The palaeomagnetic mean direction is $D_m = 86°$, $I_m = +60°$. Various palaeomagnetic parameters at specimen, sample, and site level, and for mean value of specimen poles, are given in Table I. The distributions are shown to be Fisherian by closeness-of-fit tests, with a significance level of 5% at both specimen and sample level. The number of sites is too low for reliable statistical treatment at site level.

A two-tier analysis (Watson and Irving, 1957) yields no significant between-site dispersion at a significance level of 5%, $F = 2.37$, and with respective degrees of freedom 10 (between sites) and 56 (within sites). Since between-site scatter is often attributed to

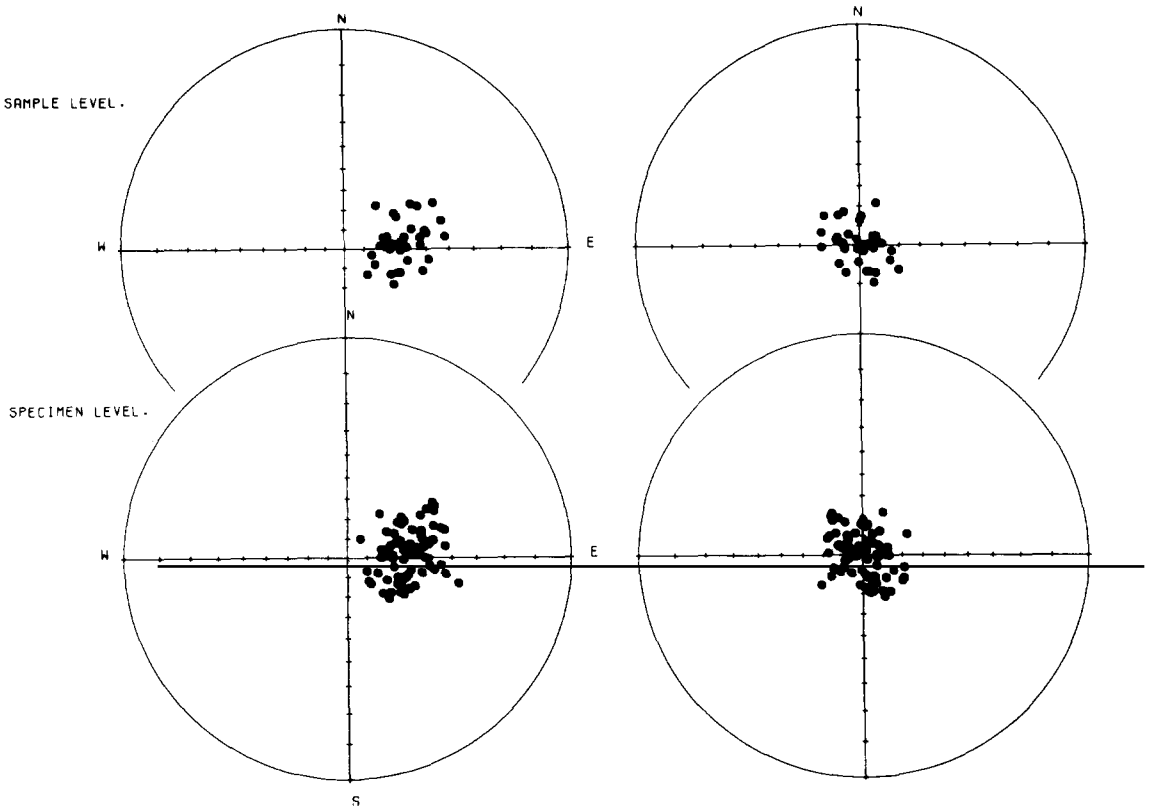


Fig. 6. Distribution of bulk directions of magnetization for the Båtsfjord dykes at sample and specimen level, before and after transformations relative to their mean values.

palaeosecular variation of the geomagnetic field, this result indicates that either (a) the dykes were intruded over a short interval of time, collecting therefore only a minor fraction of the geomagnetic secular variation, or (b) the original TRM/TCRM is completely erased by low-temperature alterations, and the secular variation is averaged out through thermochemical processes over a longer time span. In the latter model, which in the present case is suggested as by far the most likely alternative (see

TABLE I

Palaeomagnetic parameters * for the Båtsfjord dykes

	N	R	K	S_{63}	A_{95}	D_m	I_m	Pole	
								lat (°N)	long (°E)
Specimen	74	71.4	27.7	15.4	3.2	86.4	59.9	39.1	107.0
Sample	34	33.0	32.3	14.3	4.4	84.5	60.6	40.6	108.2
Site (dyke)	6	5.95	92.9	8.4	7.0	87.0	61.7	40.9	105.3
Mean value of poles (VGP_S)	74	69.2	15.3	20.8	4.4	–	–	40.2	105.6

* N = number of unit vectors; R = length of resultant; $K = (N-1)/(N-R)$, precision parameter; S_{63}, circular standard deviation; A_{95}, radius of circle of 95% confidence; D_m, I_m, declination and inclination of mean vector; pole values are relative to present grid system.

Section 5), the palaeomagnetic mean direction of the dykes may be expected to conform to that of an axial dipole field. The corresponding pole is then given by 40°N 105°E (Table I).

4.2. *The Kongsfjord dykes*

The NRM intensities are rather weak, falling in the range 10^{-5}–10^{-6} e.m.u. cm^{-3} for the majority of the 132 specimens. Although the NRM directions are scattered, their mean direction is in close agreement with that of the Båtsfjord dykes ($D_m \sim 90°$, $I_m \sim +70°$).

AF demagnetization of 40 specimens indicates predominantly low-coercive components. An intensity reduction of 50% is in most cases obtained below 200 Oe, after which the intensity as well as the directions usually exhibit erratic behaviour. Only a few specimens exhibit a regularly-decaying intensity pat-

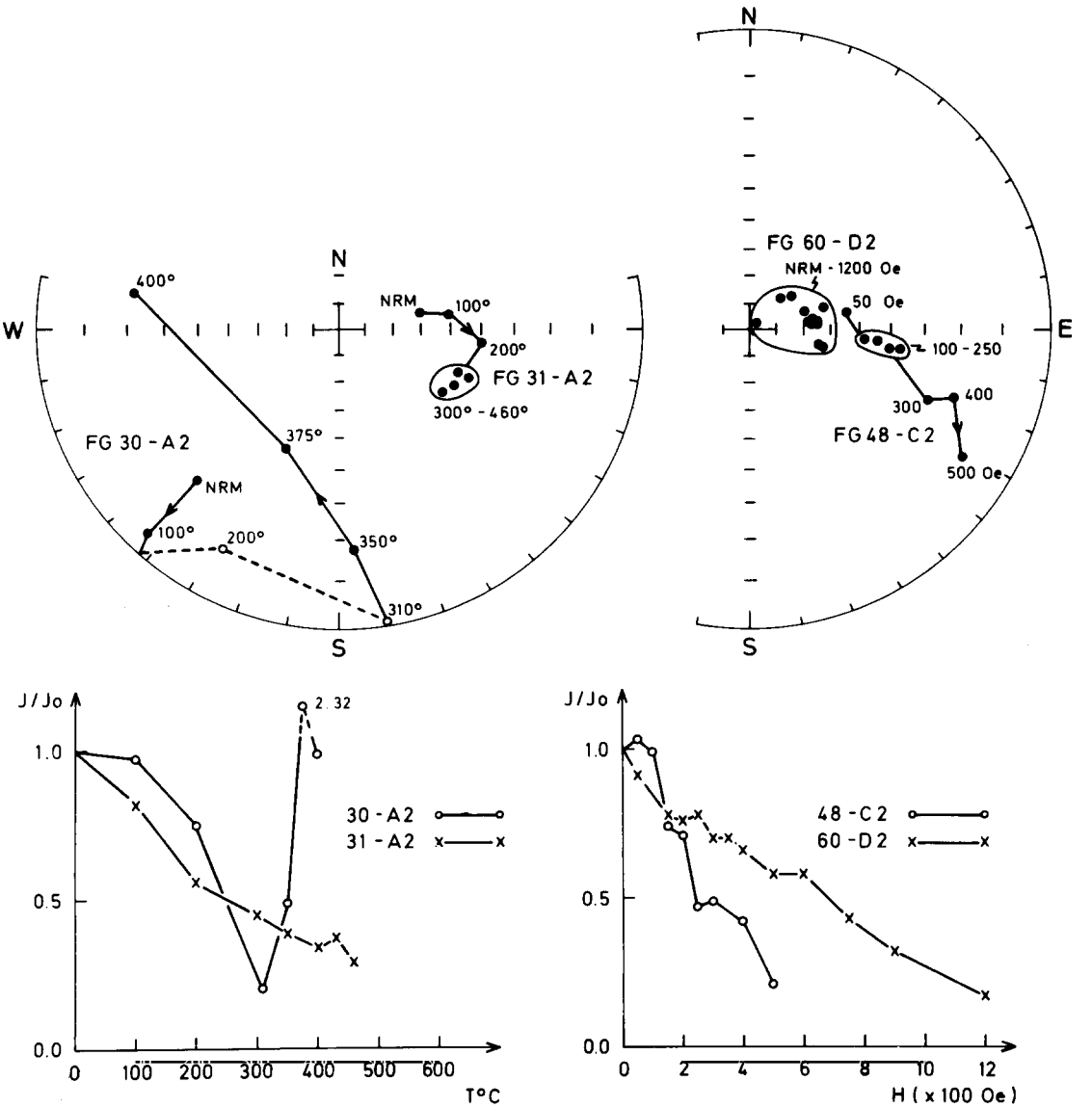


Fig. 7. AF- and thermal-demagnetization results for the Kongsfjord dykes.

tern with increasing demagnetization (cf. FG 48-C2 and FG 60-D2 of Fig. 7). The only specimen giving meaningful measurements above 500 Oe was FG 60-D2.

A typical intensity curve for thermal demagnetization is that of FG 30-A2 (Fig. 7): a smooth intensity reduction below ca. 300°C is superseded by a steady increase, or more often, by erratic oscillations of the magnetic moment. Specimen FG 31-A2 is the only specimen establishing a terminal direction on thermal treatment. Correlating these results with corresponding J_s/T data, it is suggested that the magnetization of FG 30-A2 is dominated by pyrrhotite, while FG 31-A2 shows behaviour implying a titanomagnetite/maghemite mineralogy. The sudden intensity increase in most of the thermally-demagnetized specimens may be due to the self-reversing properties of pyrrhotite (Bhimasankaram and Lewis, 1966).

It is worth noting that the few stable directions found in the Kongsfjord dyke suite are in excellent agreement with the main directional group obtained from the Båtsfjord dykes, but that in the majority of the Kongsfjord specimens no meanigful palaeomagnetic direction can be defined.

5. Discussion

As discussed earlier, the lack of statistically-significant between-site scatter in the main magnetization component of the Båtsfjord dykes is supposed to be the result of an averaging out of the geomagnetic secular variation, due to relatively longlasting thermochemical magnetization processes. Both geological and magneto-mineralogical evidence support this view. According to Roberts (1972), these dykes are syn-tectonic, generated during the Late Precambrian main deformation phase in this part of Finnmark. Roberts also suggests that the intrusive phase was followed by further low-grade metamorphic processes, which also affected the dykes. The presence of high-temperature oxidation products shows that the dykes are not quenched, and the observed extensive low-temperature alterations ($T < 250$°C) support the view that metamorphic processes took place after the intrusive phase. Thus, the original TRM/TCRM magnetization was most likely erased and replaced by TCRM components acquired over a comparatively long time interval; the evidence for a minor magnetization component antiparallel to the main component indicates that the magnetization processes lasted for a minimum of 10^5–10^6 years.

The main remanence direction does not fit in with any known Phanerozoic palaeofield for Western Europe, and it is therefore concluded that the stable magnetization of the Båtsfjord dykes ($D_m = 86°$, $I_m = +60°$) represents the relative dipole field at around the time of dyke injection (ca. 640 My B.P.). However, the pole defined by the Båtsfjord dykes (40°N 105°E; Table I) is not in agreement with other palaeopole determinations of the Late Precambrian of northwest Europe (Fig. 8). The most likely explanation of this discrepancy is that it has a geotectonic cause, and a large-scale dextral displacement along the Trollfjord–Komagelv fault zone has been proposed (Kjøde et al., 1978). This movement, which is of the order of 500–1000 km, is based on linking the Trollfjord–Komagelv fault zone with

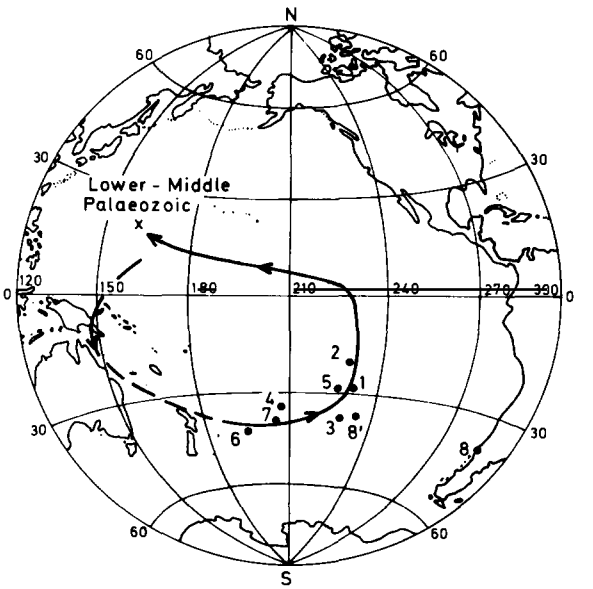


Fig. 8. The Båtsfjord pole (8) seen in conjunction with other Late Precambrian poles for NW Europe. These are: 1 = Storetvedt and Gidskehaug (1968); 2–4 = Poorter (1972); 5 = Stewart and Irving (1974); 6 = Murthy and Deutsch (1976); 7 = Murthy and Deutsch (1974). The ages of these poles are thought to range between ~950 and ~800 My B.P. Pole 8′ = Båtsfjord pole after the proposed rotation (see text). The indicated polar path for the Late Precambrian is according to Spall (1973).

the fracture system of the Svalbard/Greenland region, and is associated with an Euler pole just north of Novaya Zemlya at around 77°N 60°E. Such a displacement is in good accordance with geological considerations which predict a common depositional basin for the Barents Sea group of Varanger peninsula, the Eleonore Bay group of east Greenland and the Hecla Hoek formation of east Spitsbergen (cf. Kjøde et al., 1978, fig. 5; Harland and Gayer, 1972; Holtedahl, 1918,1932; Siedlecka and Siedlecki, 1971; Siedlecki, 1975). The principal phase of this dextral strike-slip movement is assumed to have occurred between 640 and 500 My B.P. Further discussion of this issue is given in Kjøde et al. (1978).

The systematic directional trends towards a shallow NE–SW-striking axis are also assumed to be connected with a real palaeomagnetic field axis. It is not possible to estimate a reliable mean direction from a few stable specimens only; but from the available evidence, it appears reasonable to suggest a position at $D_m \sim 45°$ and $I_m \sim 0–10°$. Such a direction will correspond well to the proposed Lower–Middle Palaeozoic pole position for Western Europe, at 20°N 160°E (Briden et al., 1970; Storetvedt, 1967, 1973). For the Båtsfjord dykes, there are obvious reasons to suggest that the generation of this shallow axis occurred during Caledonian remagnetization, after the T–K transcurrence. However, recent results (Claesson, 1978) suggest that the shallow axis may be of later origin than the principal thermotectonic event in Finnmark, which took place between 530 and 500 My B.P. (Sturt et al., 1975). Claesson reports some results from Swedish Lower Ordovician limestones, where the bulk magnetization has a high positive inclination. The associated pole (30°S 226°E) is in excellent agreement with the Late Precambrian poles of NW Europe (Fig. 8). This may well indicate that the relative polar shift for NW Europe between the Late Precambrian and the Lower–Middle Palaeozoic took place as late as the post-Lower Ordovician period. If this is the case, the prefix Lower–Middle Palaeozoic (in Fig. 8) ought to be replaced by Middle Palaeozoic, ranging from mid(?)-Ordovician to mid-Carboniferous time (Storetvedt, 1967). It should also be mentioned that the Swedish limestones give similar directional trends to those found in the present study, thus supporting the suggestion that these low-intensity components represent true palaeomagnetic fields. If so, the shallow magnetization axis of the Båtsfjord dykes must be of late Caledonian origin (post-Lower Ordovician), and the Caledonian metamorphic processes in Finnmark must still have been active a long time after the maximum activity at around 530–500 My B.P.

The rock- and palaeomagnetic results from the Kongsfjord dykes show that they have suffered an extreme degree of alteration, and that titanomagnetite has been almost entirely erased. Only three specimens give stable directions, and these are in good agreement with the main magnetic component for the Båtsfjord dykes. It is not possible from these data to conclude whether the Kongsfjord dykes are older than the Båtsfjord dykes (as suggested by Beckinsale et al., 1976), or whether they are derived from the same magmatic pulse, as suggested by Roberts (1975). More information is needed before a firm conclusion on this point can be reached. The Kongsfjord dykes must, however, have suffered a higher degree of metamorphic and hydrothermal alteration in accordance with the tectonic analysis of Roberts (1972), where he concludes that the metamorphism resulting from the D_1 deformation increases northwestwards across the Varanger peninsula.

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